
DP 503P: Basic Mechanical and Electrical Workshop

Course Code	DP 503P
Course Name	Basic Mechanical and Electrical Workshop
L-T-P-C	0-0-4-2
Prerequisites	None
Intended for	M.Sc. & Ph.D.
Distribution	Core
Approval	10th Senate (Jun 2016)

Course Contents

- **Module 1. Engineering Drawing and Solid Works:**

- Uses of Engineering Drawing and its importance, Introduction to lines, Scales and surfaces and their uses. Representation of solids using different lines. Dimensioning, unidirectional method and aligned method of dimensioning, symbolic representations of different mechanical and electrical components. Projections: Point, lines, surfaces, solids, and angle projections. Introduction to isometric views and projections.
- Introduction to CAD modeling techniques and Solid works. Sketching of different two dimensional objects using lines, curves, circle, polygons, arc, etc. Use of extrusion, revolve, extrude cut, revolve cup, linear patterns and others, in order to get three dimensional objects.

- **Module 2. Electronics workshop:**

- Introduction to CRO (Cathode ray oscilloscope)
- DSO (Digital storage oscilloscope)
- Voltage measurement and amplification using Wheat Stone Bridge
- Temperature measurement (temperature sensor)
- Binary addition and logic gates (AND, NOR, OR, XOR)
- Half and full added circuit
- Introduction to MA TLAB

Text Books

Not Available

References

Not Available